

Singular Limits for Three-Dimensional Global Minimizers of Ginzburg–Landau-Type Functionals

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Outline

- 1 Background and setting
- 2 Topological energy and network cost
- 3 Main results
- 4 Key estimates and proof strategy
- 5 Summary

The functional and its singular limit

Domain and target: Let $\Omega \subset \mathbb{R}^3$ be bounded, and let $\mathcal{N} \subset \mathbb{R}^m$ be a smooth, compact, connected manifold isometrically embedded in $\mathbb{R}^m$.

For $u \in H^1(\Omega, \mathbb{R}^m)$ and $0 < \varepsilon < 1$, set

$$E_\varepsilon(u, \Omega) := \int_{\Omega} e_\varepsilon(u) \, dx, \quad e_\varepsilon(u) := \frac{1}{2} |\nabla u|^2 + \frac{1}{\varepsilon^2} f(u), \quad (\text{GL}_\varepsilon)$$

where $f \geq 0$ and $f^{-1}(0) = \mathcal{N}$.

Critical points: If $f \in C^1$, then

$$\Delta u = \frac{1}{\varepsilon^2} Df(u).$$

Singular limit: As $\varepsilon \downarrow 0$, the potential drives u toward $\mathcal{N}$ away from a **defect set**; topology may force that set to persist.

Why the logarithmic scale?

For the classical complex model,

$$\mathcal{N} = \mathbb{S}^1, \quad f(u) = \frac{1}{4}(|u|^2 - 1)^2.$$

- In dimension two, the minimum energy in degree $d \neq 0$ is

$$\pi|d| |\log \varepsilon| + O(1)$$

[Bethuel–Brezis–Hélein, 1994; Struwe, 1994].

- In dimension three, the transverse vortex profile may extend along a curve; its leading-order energy is proportional to its length.

Natural scale in three dimensions:

$$\limsup_{\varepsilon \downarrow 0} \frac{E_\varepsilon(u_\varepsilon, \Omega)}{|\log \varepsilon|} < +\infty, \quad \dim S_* = 1.$$

Minimizers vs. Solutions

$\mathcal{N} = \mathbb{S}^1$, $n \geq 3$: logarithmic energy concentrates in codimension two.

- Uniform convergence off the singular set.
- the limit is an **integral rectifiable**, “**area-minimizing**” $(n - 2)$ -varifold.

- For $n = 3$: [Lin–Rivière, 2001];
- for all $n \geq 3$: [Bethuel–Brezis–Orlandi, 2001].
- The limit is **stationary**, possibly non-integral [Pigati–Stern, 2023].
- The limit can be **unstable, geodesic** [Colinet–Jerrard–Sternberg, 2025].

Two energy regimes

- As a one-dimensional Riemannian manifold, $\mathbb{S}^1$ is intrinsically flat.
- A general $\mathcal{N}$ may have nonzero intrinsic curvature; in dimension three, point singularities and **line defects** may coexist.

$$\sup_{0 < \varepsilon < 1} E_\varepsilon(u_\varepsilon) < +\infty.$$

For global minimizers, the limit is a **minimizing harmonic map** u_* , with

$$\dim_{\mathcal{H}} \text{sing}(u_*) \leq n - 3$$

(points when $n = 3$).

[Lin–Wang, 1999; Majumdar–Zarnescu, 2010; Contreras–Lamy–Rodiac, 2018; Contreras–Lamy, 2022].

$$\sup_{0 < \varepsilon < 1} \frac{E_\varepsilon(u_\varepsilon)}{|\log \varepsilon|} < +\infty.$$

$\pi_1(\mathcal{N}) \neq 0$ supports **codimension-two line defects**, labeled by **free homotopy classes**.

For $\mathbb{S}^1$, $\pi_1(\mathbb{S}^1) \simeq \mathbb{Z}$, so these labels reduce to integer degrees.

If $\pi_1(\mathcal{N})$ is non-Abelian, there is no single integer degree: charge addition can be **multi-valued**.

Liquid-crystal examples

Landau–de Gennes (LdG) theory uses a tensor order parameter Q ; its bulk energy wells form the vacuum manifold $\mathcal{N}$.

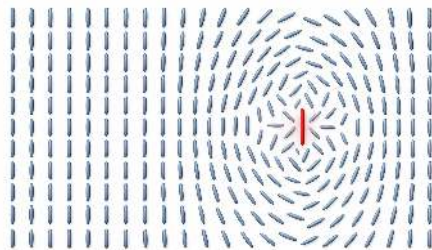
- **Uniaxial wells:** $n \sim -n$,
hence

$$\mathcal{N} \simeq \mathbb{RP}^2.$$

- **Biaxial wells:**

$$\mathcal{N} \simeq \mathbb{S}^3/Q_8.$$

- $\pi_1(\mathcal{N}) \neq 0$ produces non-trivial **line disclinations**.



ordered nematic

line disclination

What was known before

- **Two-dimensional vortices:**

- Convergence and defect-core analysis for uniaxial Landau–de Gennes [Canevari, 2015];
- For general $\mathcal{N}$, compactness away from finitely many vortex points and renormalized-energy selection of their locations [Monteil–Rodiac–Van Schaftingen, 2021, 2022].

- **Three-dimensional interior networks:** Results for $\mathbb{RP}^2$ [Canevari, 2017] and $\mathbb{S}^3/Q_8$ [Wang–Zhang, 2024].

- **All dimensions:** Γ -convergence when $\pi_1(\mathcal{N})$ is Abelian [Canevari–Orlandi, 2021].

Three open tasks in dimension three.

- Prove compactness at logarithmic scale and identify a **finite weighted network of straight segments**.
- Determine the **boundary endpoints** and prove uniform estimates up to $\partial\Omega$.
- Establish **weighted-length minimality** of the labeled network.

Assumptions in dimension three

Domain: $\Omega \subset \mathbb{R}^3$ is bounded and $C^{2,1}$, and $\overline{\Omega}$ is **strongly convex** at every boundary point; hence $\partial\Omega$ is simply connected.

Vacuum manifold:

- $f \in C^\infty(\mathbb{R}^m, [0, +\infty))$;
- $\mathcal{N} = f^{-1}(0)$ is a **compact, connected, smooth manifold without boundary**;
- $\#\pi_1(\mathcal{N}) < +\infty$.

Coercivity and non-degeneracy:

$$\liminf_{|y| \rightarrow \infty} f(y) > 0, \quad y \cdot Df(y) \geq 0,$$

for $|y|$ large, and

$$D^2f(x)[v, v] > 0 \quad \text{for } 0 \neq v \perp T_x\mathcal{N}.$$

Finite $\pi_1(\mathcal{N})$ yields the uniform **L^1 gradient estimate** used to prove $W^{1,q}$ bounds. Without finiteness this mechanism may fail. For $\mathcal{N} = \mathbb{S}^1$, even

$$E_\varepsilon = o(|\log \varepsilon|)$$

need not imply local $W^{1,1}$ bounds or L^1_{loc} compactness [Bethuel–Brezis–Orlandi, 2001, Remark III.4].

Boundary data and prescribed charges

Let $\mathcal{A} = \{a_1, \dots, a_k\} \subset \partial\Omega$. The maps $g_\varepsilon \in C^2(\partial\Omega, \mathbb{R}^m)$ regularize these charges at scale ε , with

$$g_\varepsilon \rightarrow g \quad \text{in } C_{\text{loc}}^2(\partial\Omega \setminus \mathcal{A}), \quad g \in C^2(\partial\Omega \setminus \mathcal{A}, \mathcal{N}).$$

- **Scale-invariant control:**

$$|D_{\partial\Omega}^j g_\varepsilon(x)| \leq M_0 (\max\{\text{dist}(x, \mathcal{A}), \varepsilon\})^{-j}, \quad j = 0, 1, 2;$$

- **Exact vacuum trace:** $g_\varepsilon = g \in \mathcal{N}$ away from the radius- ρ cores whenever $\varepsilon < \rho$;
- **Non-trivial charge:** every loop $g_\varepsilon|_{\partial\Omega \cap \partial B_\rho(a_i)}$ is non-trivial in $\mathcal{N}$;
- **Controlled angular core:** After flattening and rescaling, each core has a uniformly controlled angular profile.

Regularizing each boundary core

After flattening near a_i and rescaling by ε , the angular-core hypothesis prescribes

$$g_\varepsilon(x) = g_\varepsilon^0\left(\frac{\Phi_0(R(x - a_i))}{\varepsilon}\right), \quad g_\varepsilon^0(y) = h_\varepsilon\left(\frac{y'}{|y'|}\right) \chi(|y'|).$$

- $\Phi_0(0) = 0$, $D\Phi_0(0) = \text{Id}$, while $h_\varepsilon : \mathbb{S}^1 \rightarrow \mathcal{N}$ is uniformly C^2 .
- This is the hypothesis of [Lin–Rivière, 1999, p. 243(iii)]; it makes each g_ε a smooth ε -core regularization.
- In contrast with the p -harmonic model [Bulanyi–Van Schaftingen–Van Vaerenbergh, 2025], the potential and the ambient-valued core create **additional boundary errors**.

The angular structure controls the **boundary stress–energy error** down to scale ε .

The variational problem

u_ε is a **global minimizer** with trace g_ε :

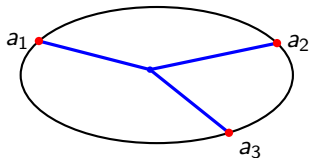
$$u_\varepsilon \in \operatorname{argmin} \{E_\varepsilon(v, \Omega) : \operatorname{tr} v = g_\varepsilon\}.$$

Normalized energy measure:

$$\mu_\varepsilon := \frac{e_\varepsilon(u_\varepsilon)}{|\log \varepsilon|} dx.$$

Weak-* limits of μ_ε record the leading-order energy concentration.

The problem is to identify the support of μ_* , the limit map u_* , and the **network selected by global minimality**.



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Free homotopy and line defects

Two continuous loops $\ell_1, \ell_2 \in C^0(\mathbb{S}^1, \mathcal{N})$ satisfy $\ell_1 \sim \ell_2$ if there exists

$$G \in C^0([0, 1] \times \mathbb{S}^1, \mathcal{N}), \quad G(0, \cdot) = \ell_1, \quad G(1, \cdot) = \ell_2.$$

The quotient $[\mathbb{S}^1, \mathcal{N}]$ is the set of free homotopy classes.

Algebraic description: Choosing a base point gives

$$[\mathbb{S}^1, \mathcal{N}] \longleftrightarrow \{\text{conjugacy classes of } \pi_1(\mathcal{N})\}.$$

Label of a line defect: For a disk D transverse to the defect, assume $u|_{\partial D} \in C^0(\partial D, \mathcal{N})$; then define

$$\sigma = [u|_{\partial D}]_{\mathcal{N}}.$$

The class σ is unchanged as D slides through a regular tube that avoids the other defects.

The cost of a homotopy class

Least loop energy in a fixed class:

$$E_{\min}(\sigma) := \inf \left\{ \frac{1}{2} \int_{\mathbb{S}^1} |u'|^2 d\mathcal{H}^1 : u \in H^1(\mathbb{S}^1, \mathcal{N}), [u]_{\mathcal{N}} = \sigma \right\}.$$

$$|\sigma|_* := \inf \left\{ \sum_{i=1}^k E_{\min}(\sigma_i) : \begin{array}{l} k \in \mathbb{N}, \sigma_i \in [\mathbb{S}^1, \mathcal{N}], \\ \sigma \in \sigma_1 + \cdots + \sigma_k \end{array} \right\}.$$

The sum is understood in the iterated multi-valued sense.

The assumption $\#\pi_1(\mathcal{N}) < \infty$ gives a **positive energy gap**:

$$|\sigma|_* = 0 \iff \sigma = 0, \quad |\sigma|_* \geq c_0 > 0 \quad (\sigma \neq 0),$$

while the definition makes $|\cdot|_*$ subadditive under charge merging.

Junction topology versus geometry

For free homotopy classes, addition is induced by multiplication of conjugacy classes:

$$\alpha + \beta = \{\text{conjugacy classes of } ab : a \in \alpha, b \in \beta\}.$$

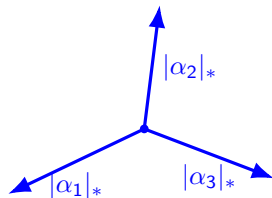
Topological compatibility: For incident labels

$\alpha_1, \dots, \alpha_\ell$,

$$0 \in \alpha_1 + \dots + \alpha_\ell.$$

Force balance: After stationarity is established, the tangent directions satisfy

$$\sum_{j=1}^{\ell} |\alpha_j|_* v_j = 0.$$



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Normalized energy measures

Energy bounds: Global minimality and the boundary construction give

$$E_\varepsilon(u_\varepsilon, \Omega) \leq C(|\log \varepsilon| + 1), \quad \|u_\varepsilon\|_{L^\infty(\Omega)} \leq C.$$

Normalize the energy density: Set

$$\mu_\varepsilon := \frac{e_\varepsilon(u_\varepsilon)}{|\log \varepsilon|} dx, \quad \mu_\varepsilon(\overline{\Omega}) = \frac{E_\varepsilon(u_\varepsilon, \Omega)}{|\log \varepsilon|} \leq C \left(1 + \frac{1}{|\log \varepsilon|}\right) \leq C'$$

for $0 < \varepsilon \leq \frac{1}{2}$. Thus $\{\mu_\varepsilon\}$ has uniformly bounded total mass along any vanishing sequence.

Weak-* compactness: Along a sequence $\varepsilon_i \downarrow 0$, for some $\mu_* \in \mathcal{M}^+(\overline{\Omega})$,

$$\mu_{\varepsilon_i} \xrightarrow{*} \mu_* \quad \text{in } (C^0(\overline{\Omega}))'.$$

The concentration set is the support of the limiting measure:

$$S_* := \text{supp } \mu_*.$$

Compactness away from the network

The interior concentration set is **1-rectifiable** and has finite length:

$$\mathcal{H}^1(S_* \cap \Omega) < +\infty.$$

Moreover, there exists

$$u_* \in H^1(\Omega \setminus S_*, \mathcal{N})$$

which **locally minimizes the Dirichlet energy**, and

$$u_{\varepsilon_i} \rightarrow u_* \quad \text{strongly in } H_{\text{loc}}^1(\Omega \setminus S_*, \mathbb{R}^m).$$

Residual point singularities: The set $S_0 := \text{sing}(u_*)$ is locally finite in $\Omega \setminus S_*$.
On $\Omega \setminus (S_* \cup S_0)$,

$$u_* \in C^\infty(\Omega \setminus (S_* \cup S_0), \mathcal{N}), \quad u_{\varepsilon_i} \rightarrow u_* \quad \text{in } C_{\text{loc}}^0.$$

The limiting network

There is a finite family of closed line segments $L_j \subset \overline{\Omega}$, with pairwise disjoint relative interiors, such that

$$S_* = \bigcup_{j \in J} L_j.$$

For $x \in \text{Int } L_j$, choose a transverse disk D with $D \cap S_* = \{x\}$ and $\partial D \cap S_0 = \emptyset$. Define

$$\alpha_j := [u_*|_{\partial D}]_{\mathcal{N}} \neq 0, \quad \sigma_j := |\alpha_j|_* = |[u_*|_{\partial D}]_{\mathcal{N}}|_* > 0.$$

Then the limiting measure is the weighted length measure

$$\mu_* = \sum_{j \in J} \sigma_j \mathcal{H}^1 \llcorner L_j.$$

L_j : geometry α_j : meridian class σ_j : energy per unit length

Density quantization and force balance

The meridian class α_j , and hence the line tension σ_j , is independent of the choices of $x \in \text{Int } L_j$ and the admissible transverse disk D .

Pointwise quantization: The one-dimensional density is **exactly the transverse topological cost**:

$$\Theta^1(\mu_*, x) := \lim_{r \rightarrow 0^+} \frac{\mu_*(\overline{B}_r(x))}{2r} = \sigma_j = |[u_*|_{\partial D}]_{\mathcal{N}}|_*.$$

Junction law: At an interior junction x , let v_j be the outward unit tangent of the j -th incident segment. Stationarity then yields

$$\sum_{j \in J_x} \sigma_j v_j = 0.$$

Boundary anchoring

Strong convexity and the prescribed charges give

$$S_* \cap \partial\Omega = \mathcal{A}, \quad S_* \subset \text{Conv}(\mathcal{A}).$$

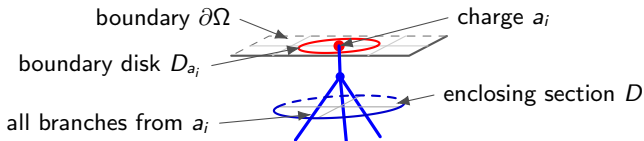
Thus there is no concentration on $\partial\Omega \setminus \mathcal{A}$; endpoints lie in $\mathcal{A}$ or at interior branching points.

Matching the boundary charge: Choose $D_{a_i} \subset \partial\Omega$ around a_i and an interior section $\bar{D} \subset \Omega$ enclosing all branches issuing from a_i , with

$$\partial D \cap (\text{Conv}(\mathcal{A}) \cup S_0) = \emptyset.$$

A regular annulus joins ∂D_{a_i} to ∂D ; hence

$$[g|_{\partial D_{a_i}}]_{\mathcal{N}} = [u_*|_{\partial D}]_{\mathcal{N}}.$$



Uniform estimates up to the boundary

For any $q \in (1, 2)$, the following estimates hold uniformly for $0 < \varepsilon < 1$:

$$\|\nabla u_\varepsilon\|_{L^q(\Omega)} \leq C_q, \quad \frac{1}{\varepsilon^2} \int_{\Omega} f(u_\varepsilon) \, dx \leq C.$$

Both estimates hold on all of Ω , including neighborhoods of $\partial\Omega$.

- **Interior precedents:** Complex GL [Bourgain–Brezis–Mironescu, 2004; Bethuel–Orlandi–Smets, 2005] and Landau–de Gennes [Fu–Wang–W., 2026].
- **Boundary statement:** The $W^{1,q}$ and potential bounds are uniform on all of Ω , including neighborhoods of $\partial\Omega$.
- **Proof and sharpness:** Boundary partial regularity replaces the missing p -harmonic control of the potential. The endpoint $q = 2$ fails in general, while the potential energy remains on the optimal $O(1)$ scale.

The admissible class $\mathcal{C}(U, h)$

Competitor data: Let $U \subset \mathbb{R}^3$ be bounded Lipschitz, $h \in H^{\frac{1}{2}}(\partial U, \mathcal{N})$, and I be finite. Write

$$\Gamma = \{(K_i, \beta_i)\}_{i \in I}, \quad S_\Gamma := \bigcup_{i \in I} K_i.$$

Here $K_i \subset \overline{U}$ is a closed segment and $\beta_i \in [\mathbb{S}^1, \mathcal{N}]$.

Geometry:

- finitely many positive-length closed segments with pairwise disjoint relative interiors;
- every endpoint lies on ∂U or is an interior branching point;
- $S_\Gamma \cap \partial U$ is finite.

Realization by a map:

There exist a locally finite set $F \subset U \setminus S_\Gamma$ and a map

$$v \in H_{\text{loc}}^1(\overline{U} \setminus S_\Gamma, \mathcal{N}) \cap C^0(U \setminus (S_\Gamma \cup F), \mathcal{N})$$

with trace h in the local trace sense. For any $x \in \text{Int } K_i$ and any closed transverse disk $\overline{D} \subset U$ centered at x such that $D \cap S_\Gamma = \{x\}$ and $\partial D \cap F = \emptyset$,

$$[v|_{\partial D}]_{\mathcal{N}} = \beta_i.$$

Weighted-length minimality

For $\Gamma = \{(K_i, \beta_i)\}_{i \in I}$, set

$$\mathbb{M}(\Gamma) := \sum_{i \in I} |\beta_i|_* \mathcal{H}^1(K_i).$$

Set

$$\Gamma_* := \{(L_j, \alpha_j) : j \in J\}.$$

Then $\Gamma_* \in \mathcal{C}(\Omega, g)$, and

$$\mathbb{M}(\Gamma_*) \leq \mathbb{M}(\Gamma) \quad \text{for any } \Gamma \in \mathcal{C}(\Omega, g).$$

The comparison class contains **map-realizable labeled networks**, not arbitrary unions of segments.

The closest analogue: p -harmonic maps as $p \uparrow 2$

Set $\mathcal{E}_p(v) := \frac{1}{p} \int_{\Omega} |\nabla v|^p dx$. The two singular perturbations compare as follows.

	p -harmonic limit	present GL limit
Maps	$v_p : \Omega \rightarrow \mathcal{N}$	$u_{\varepsilon} : \Omega \rightarrow \mathbb{R}^m$
Scale	$\mathcal{E}_p(v_p) = O\left(\frac{1}{2-p}\right)$	$E_{\varepsilon}(u_{\varepsilon}) = O(\log \varepsilon)$
Boundary	fixed singular $\mathcal{N}$ -valued trace g	ε -dependent, ambient-valued regularization g_{ε}

Common limit: If $\#\pi_1(\mathcal{N}) < \infty$, both theories give $W^{1,q}$ control for $q < 2$ and convergence away from a **weighted one-dimensional network**. Under the corresponding convexity and boundary hypotheses, that network is a finite union of straight segments and a **homotopical Plateau minimizer**.

What does not transfer: The network and Plateau arguments are adapted from the p -harmonic theory, but its $W^{1,q}$ proof does not apply directly. The potential $\varepsilon^{-2}f(u_{\varepsilon})$ and the boundary cores require the Fu–Wang–Wang scheme, potential estimates, and **boundary partial regularity**.

[Bulanyi–Van Schaftingen–Van Vaerenbergh, 2025]

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The two-dimensional line cost

Let $u \in H^1(B_r^2, \mathbb{R}^m)$ and $g := u|_{\partial B_r^2} \in H^1(\partial B_r^2, \mathbb{R}^m)$. Assume $\text{dist}(g, \mathcal{N}) < \delta_{\mathcal{N}}$, and set

$$\sigma := [\Pi_{\mathcal{N}} \circ g]_{\mathcal{N}}.$$

$$E_{\varepsilon}(u, B_r^2) + CrE_{\varepsilon}(u, \partial B_r^2) \geq |\sigma|_* \log \frac{r}{\varepsilon} - C.$$

If $g \in H^1(\partial B_r^2, \mathcal{N})$, one can choose $\text{tr } u_{\varepsilon} = g$ with

$$E_{\varepsilon}(u_{\varepsilon}, B_r^2) \leq |\sigma|_* \log \frac{r}{\varepsilon} + C \left(r \|\nabla_{\top} g\|_{L^2(\partial B_r^2)}^2 + 1 \right).$$

The coefficient of the logarithmic term is therefore the exact line tension: $|\sigma|_*$.

Interior monotonicity

For an interior point x , set

$$\theta_\varepsilon(u; x, r) := \frac{1}{r} \int_{B_r(x)} e_\varepsilon(u).$$

For Euler–Lagrange solutions, θ_ε is **monotone** in the interior:

$$\begin{aligned} \theta_\varepsilon(u; x, t) - \theta_\varepsilon(u; x, s) = \int_s^t \left[\frac{1}{\rho} \int_{\partial B_\rho(x)} |\partial_r u|^2 d\mathcal{H}^2 \right. \\ \left. + \frac{2}{\varepsilon^2 \rho^2} \int_{B_\rho(x)} f(u) \right] d\rho \geq 0. \end{aligned}$$

Passage to the limit. If $\mu_{\varepsilon_i} \xrightarrow{*} \mu_*$, then

$$r \mapsto \frac{\mu_*(\overline{B}_r(x))}{2r} \text{ is non-decreasing, } \Theta^1(\mu_*, x) := \lim_{r \downarrow 0} \frac{\mu_*(\overline{B}_r(x))}{2r}.$$

For an interior local minimizer with $\|u\|_{L^\infty(B_{2r}(x))} \leq M$, there exist $\delta \in (0, 1)$ and $C = C(f, \mathcal{N}, M) > 0$ such that

$$\varepsilon < \delta r, \quad \theta_\varepsilon(u; x, r) \leq \delta \log \frac{r}{\varepsilon} \implies \theta_\varepsilon\left(u; x, \frac{r}{2}\right) \leq C.$$

At the limit, the density gap reads

$$\mu_*(\overline{B}_r(x)) < \delta r \implies \mu_*(B_{\frac{r}{2}}(x)) = 0.$$

Consequently, every $x \in S_* \cap \Omega$ carries **uniformly positive density**.

Rectifiability and straight segments

Density detection: Clearing-out and monotonicity give

$$S_* \cap \Omega = \{x : \Theta^1(\mu_*, x) \geq \delta_0\}.$$

Stationary varifold:

The limiting stress–energy tensor induces a **stationary 1-varifold** V in the interior, with approximate unit tangent τ :

$$\delta V(\xi) = \int_{S_* \cap \Omega} (\tau \otimes \tau) : D\xi \, d\mu_* = 0 \quad \forall \xi \in C_c^1(\Omega, \mathbb{R}^3).$$

Structure:

clearing-out + monotonicity + stress–energy compactness

$\implies$ rectifiable stationary 1-varifold,

cylindrical quantization + Allard–Almgren

$\implies$ locally finite straight network.

Density quantization in a cylinder

A thin cylinder $\Lambda_{\delta,1} := B_\delta^2 \times (-1,1)$ crosses a single segment (after blow-up). Its lateral trace has class σ , and the excess energy is $\eta \ll 1$. Then

$$E_{\bar{\varepsilon}}(u_\varepsilon, \Lambda_{\delta,1}) = 2|\sigma|_* \log \frac{\delta}{\bar{\varepsilon}} + O\left((\delta + \eta + \delta^{-1}\eta) \log \frac{\delta}{\bar{\varepsilon}}\right) + O(1).$$

- The lower bound follows by slicing into transverse disks.
- The upper bound uses Luckhaus interpolation and an optimal core vortex.
- First let $\bar{\varepsilon} \downarrow 0$, and then $\eta, \delta \downarrow 0$.

$$\Theta^1(\mu_*, x) = |\sigma|_* \quad \text{for } \mathcal{H}^1\text{-a.e. } x \in S_*.$$

Boundary monotonicity

For $x_0 \in \partial U$, set

$$U_r(x_0) := U \cap B_r(x_0), \quad T_r^U(x_0) := \partial U \cap B_r(x_0), \quad \theta_\varepsilon^U(u; x_0, r) := \frac{1}{r} \int_{U_r(x_0)} e_\varepsilon(u).$$

Let U be bounded and $C^{2,1}$. Assume that u solves the Euler–Lagrange equation in $U_{2r}(x_0)$ and $u = g$ on $T_{2r}^U(x_0)$, where

$$g \in C^1(T_{2r}^U(x_0), \mathcal{N}), \quad r \|\nabla_\top g\|_{L^\infty} \leq M_0,$$

and $\theta_\varepsilon^U(u; x_0, 2r) \leq M_1$. For $0 < s < t < r$,

$$\theta_\varepsilon^U(u; x_0, t) - \theta_\varepsilon^U(u; x_0, s) + \alpha(M_0, M_1, r)(t - s) \geq \int_s^t \mathcal{R}_\varepsilon(\rho) \, d\rho \geq 0,$$

where

$$\mathcal{R}_\varepsilon(\rho) := \frac{1}{\rho} \int_{U \cap \partial B_\rho(x_0)} |\partial_r u|^2 \, d\mathcal{H}^2 + \frac{2}{\varepsilon^2 \rho^2} \int_{U_\rho(x_0)} f(u),$$
$$\alpha(M_0, M_1, r) := \frac{CM_0}{r} \left(M_0 + M_1^{\frac{1}{2}} \right) + C(M_0^2 + M_1).$$

Let U be bounded and $C^{2,1}$, and let $g \in H^1(T_{2r}^U(x_0), \mathbb{R}^m)$. Assume that u is a local minimizer in $U_{2r}(x_0)$ up to $T_{2r}^U(x_0)$, with trace g , and

$$\|u\|_{L^\infty(U_{2r})} + \|g\|_{L^\infty(T_{2r}^U)} + E_\varepsilon(g, T_{2r}^U) \leq M.$$

Here g may be **ambient-valued**; no $\mathcal{N}$ -valued trace assumption is required. Then there exist $\delta \in (0, 1)$ and $C > 0$ such that

$$\varepsilon < \delta r, \quad \theta_\varepsilon^U(u; x_0, r) < \delta \log \frac{r}{\varepsilon} \implies \theta_\varepsilon^U\left(u; x_0, \frac{r}{2}\right) \leq C.$$

Proof.

- Fubini selects $\rho \in (\frac{r}{2}, r)$ with controlled energy on the spherical side $U \cap \partial B_\rho(x_0)$.
- The cap $T_\rho^U(x_0)$ contributes only $O(1)$ through the boundary energy of g .
- Since $U_\rho(x_0)$ is bi-Lipschitz equivalent to a ball, a Luckhaus extension competitor, together with minimality, gives the bound.

Boundary partial regularity

Regular boundary patch: Away from $\mathcal{A}$, the trace satisfies $g_\varepsilon = g \in \mathcal{N}$ and

$$r \|\nabla_{\top} g\|_{L^\infty(T_{2r}^U)} + r^2 \|D_{\top}^2 g\|_{L^\infty(T_{2r}^U)} \leq M.$$

Assume also $\|u\|_{L^\infty(U_{2r})} + \theta_\varepsilon^U(u; x_0, 2r) \leq M$.

Pointwise estimate: For $\varepsilon < \delta r$,

$$(\delta r)^2 \|e_\varepsilon(u)\|_{L^\infty(U_{\delta r}(x_0))} \leq C, \quad \|f(u)\|_{L^\infty(U_{\delta r}(x_0))} \leq C\varepsilon^4 r^{-4}.$$

How the good scale is found:

- Almost-monotonicity and a dyadic pigeonhole argument give a scale s_0 with small density drop and small $s_0 \|\nabla_{\top} g\|_\infty$.
- After flattening, a blow-up limit is a **0-homogeneous** minimizing $\mathcal{N}$ -valued map on a half-ball with constant trace; hence by [Schoen–Uhlenbeck, 1983] it is constant.
- Boundary ε -regularity then gives the two pointwise bounds.

Why convexity matters

Fix $x_0 \in \partial\Omega \setminus \mathcal{A}$. On a small boundary patch, $g_\varepsilon = g \in \mathcal{N}$ and

$$\frac{1}{|\log \varepsilon|} \int_{T_{2r}^\Omega(x_0)} |\nabla_\top g_\varepsilon|^2 d\mathcal{H}^2 \longrightarrow 0.$$

Set $T_\varepsilon^{ij}(u) := e_\varepsilon(u)\delta_{ij} - \partial_i u \cdot \partial_j u$. If

$$\frac{T_\varepsilon(u_\varepsilon)}{|\log \varepsilon|} dx \xrightarrow{*} T_*,$$

then

$$\int_{\Omega_r(x_0)} \langle D\xi, dT_* \rangle \geq 0 \quad \text{whenever } \xi \cdot \nu \leq 0 \text{ on } T_r^\Omega(x_0).$$

Barrier argument. Boundary clearing-out gives a **positive density gap**. Together with interior rectifiability and the one-sided variation, strong convexity excludes contact at x_0 [White, 2010]. Hence

$$S_* \cap (\partial\Omega \setminus \mathcal{A}) = \emptyset.$$

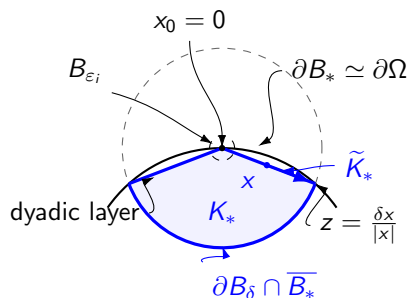
Identifying the boundary endpoints

- ① **No spurious boundary points:** The preceding first-variation and barrier arguments give $S_* \cap \partial\Omega \subset \mathcal{A}$.
- ② **Every prescribed charge occurs:** If $a_i \notin S_*$, boundary clearing-out and partial regularity produce a regular collar. The non-trivial loop $g|_{\partial D_{a_i}}$ can then be pushed to an interior loop bounding a disk on which u_* is continuous and $\mathcal{N}$ -valued—a contradiction.
- ③ **The network stays in the convex hull:** At a point maximizing the distance from $\text{Conv}(\mathcal{A})$, the force-balance law requires every branch to be tangent to the corresponding level set. Stationarity then rules out such a component.

$$S_* \cap \partial\Omega = \mathcal{A}, \quad S_* \subset \text{Conv}(\mathcal{A}).$$

The regular boundary collar also identifies each boundary charge with the class of an enclosing interior section.

The boundary cone construction



Local model. After a fixed local $C^{2,1}$ change of coordinates, move x_0 to 0 and use

$$B_* := \{y : |y + e_3| < 1\},$$

tangent to $\partial\Omega$. For a good radius δ , set

$$C_\delta := \partial B_\delta \cap \overline{B_*},$$

$$K_* := \left\{ y \in B_* \cap B_\delta : \frac{\delta y}{|y|} \in C_\delta \right\},$$

$$\tilde{K}_* := \partial K_* \setminus \partial B_\delta.$$

Radial extension

$$v_i(x) := u_{\varepsilon_i} \left(\frac{\delta x}{|x|} \right),$$

$$x \in K_* \setminus B_{\varepsilon_i}.$$

Equality forces radial branches

For radii avoiding branching points, define the total spherical density

$$h(r) := \int_{S_* \cap \partial B_r} \Theta^1(\mu_*, x) d\mathcal{H}^0(x).$$

Cone comparison: The dyadic layer is negligible at logarithmic scale. At a good radius δ , comparison with the radial extension gives

$$\int_0^\delta h(r) dr \leq \delta h(\delta).$$

A minimizing radius: Choose δ where h attains its minimum. Since $h(r) \geq h(\delta)$ on $(0, \delta)$, the reverse inequality holds. Consequently,

$$\frac{\mu_*(\Omega_r)}{r} = h(\delta) \quad (0 < r < \delta).$$

Equality in the monotonicity formula gives $(x - x_0)^\perp = 0$ μ_* -a.e.; hence only **finitely many radial straight branches** meet x_0 .

Two regularity scales

Let $U \subset \mathbb{R}^3$ and $x \in U$. Two scales encode different notions of regularity.

Pointwise regularity scale

$$r(u; x) := \sup \{ 0 \leq s \leq 1 : s^2 \|e_\varepsilon(u)\|_{L^\infty(U_s(x))} \leq 1 \}.$$

Set $\text{Bad}_I(u; r) := \{x : r(u; x) < r\}$.

Energy regularity scale For a fixed threshold $\Lambda > 0$, define

$$r^\Lambda(u; x) := \sup \{ 0 \leq s \leq 1 : \theta_\varepsilon^U(u; x, s) \leq \Lambda \},$$

and

$$\text{Bad}_{II}(u; r, \Lambda) := \{x : r^\Lambda(u; x) < r\}.$$

Clearing-out converts control of Bad_{II} into a volume estimate for Bad_I .

Three geometric regions

Boundary case: Once $\mathcal{S}_* \cap \partial\Omega = \mathcal{A}$ is known, choose finitely many boundary patches, each containing at most one charge $x_0 \in \mathcal{A}$. At scale r , cover

$$\text{Bad}_{\text{II}}(u_\varepsilon; r, \Lambda) \cup \{x_0\}$$

by at most Cr^{-1} balls of radius r .

The domain then splits into three regions:

- 1 **Near the regular boundary:** The trace is smooth and $\mathcal{N}$ -valued, so boundary partial regularity gives a uniform scale.
- 2 **In the interior:** Use the interior covering and interior regularity estimates. We actually use arguments in [Fu–Wang–W., 2025, 2026].
- 3 **Inside the covering balls:** Their total volume is $Cr^{-1}r^3 = Cr^2$.

Distribution estimate: Fix $q \in (1, 2)$ and choose $\gamma \in (\frac{q}{2}, 1)$. Then

$$\mathcal{L}^3(\{x : r(u_\varepsilon; x) < \delta R\}) \leq CR^{2\gamma}, \quad \nabla u_\varepsilon \in L^{2\gamma, \infty} \hookrightarrow L^q \quad (q < 2\gamma).$$

Controlling the potential energy

Pointwise decay. Partial regularity gives

$$f(u_\varepsilon) \leq C\varepsilon^4 r^{-4}.$$

Split the integral into

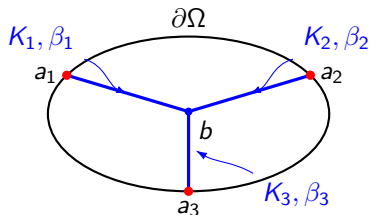
- the regular region, using pointwise decay;
- the bad region, using the covering-volume bound;
- boundary collars, using partial regularity.

For an $O(\varepsilon)$ core around a one-dimensional network, multiscale summation gives

$$\int_{\Omega} f(u_\varepsilon) \, dx \leq C\varepsilon^2.$$

Passing minimality to the limit

Fix an arbitrary $\Gamma = \{(K_i, \beta_i)\} \in \mathcal{C}(\Omega, g)$ with a realizing map v .



Recovery sequence

$$w_i = g_{\varepsilon_i} \quad \text{on } \partial\Omega,$$

$$\limsup_{i \rightarrow \infty} \frac{E_{\varepsilon_i}(w_i, \Omega)}{|\log \varepsilon_i|} \leq \mathbb{M}(\Gamma).$$

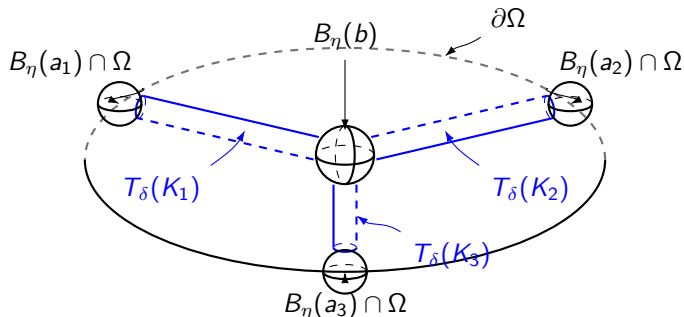
Global minimality: Since u_{ε_i} is a global minimizer,

$$E_{\varepsilon_i}(u_{\varepsilon_i}) \leq E_{\varepsilon_i}(w_i).$$

Since $\mu_*(\overline{\Omega}) = \mathbb{M}(\Gamma_*)$, passage to the limit gives

$$\mathbb{M}(\Gamma_*) \leq \mathbb{M}(\Gamma).$$

Recovery near cylinders and balls



- **Cylinders:** insert optimal class- β_i vortex profiles.
- **Balls:** glue the tube caps, **match** g_{ε_i} , and retain v outside.

$$\limsup_{i \rightarrow \infty} \frac{E_{\varepsilon_i}(w_i, \Omega)}{|\log \varepsilon_i|} \leq \sum_i |\beta_i|_* \mathcal{H}^1(K_i) = \mathbb{M}(\Gamma).$$

Outline

- 1 Background and setting
- 2 Topological energy and network cost
- 3 Main results
- 4 Key estimates and proof strategy
- 5 Summary

Two scales in the proof

Compactness scale:

$E_\varepsilon \lesssim |\log \varepsilon| \implies$ monotonicity and clearing-out $\implies$ rectifiable stationary limit.

Topological scale:

two-dimensional line cost $\implies$ density quantization $\implies$ finite weighted network.

Minimality and uniform estimates:

recovery in cylinders and balls $\implies \mathbb{M}(\Gamma_*) \leq \mathbb{M}(\Gamma)$,

regular scales and multiscale covers $\implies W^{1,q}$ and potential bounds.

Conclusion

For three-dimensional global minimizers with finite $\pi_1(\mathcal{N})$, the normalized energy converges to

$$\mu_* = \sum_{j \in J} |\alpha_j|_* \mathcal{H}^1 \llcorner L_j, \quad S_* = \bigcup_{j \in J} L_j.$$

The network is finite, its densities are quantized, and

$$S_* \cap \partial\Omega = \mathcal{A}, \quad S_* \subset \text{Conv}(\mathcal{A}).$$

Uniformly for every $q < 2$,

$$\|\nabla u_\varepsilon\|_{L^q(\Omega)} \leq C_q, \quad \frac{1}{\varepsilon^2} \int_{\Omega} f(u_\varepsilon) \, dx \leq C.$$

Finally, Γ_* solves the homotopical Plateau problem

$$\mathbb{M}(\Gamma_*) \leq \mathbb{M}(\Gamma) \quad \text{for every } \Gamma \in \mathcal{C}(\Omega, g).$$

Topology fixes the admissible charges; global minimality selects the geometry.

Thank you for your attention!